

MedCheck - AI

Multi-Modal DL System for Real-Time Pharmaceutical Authentication using Smartphone

Introduction

Counterfeit medications constitute 10–15% of global drug supply, causing over 1 million deaths annually with \$200 billion economic impact (WHO). Traditional chemical verification methods (\$10K–\$100K) are impractical for end users. MedCheck-AI presents a novel multi-modal deep learning framework combining packaging analysis, NLP-based text verification, and physical pill characterization using standard smartphone cameras. ResNet-50 achieved 95.38% accuracy with 100% counterfeit recall and 1.8s inference time. SMOTE addressed extreme 11.81:1 class imbalance with statistical validation (Cohen’s Kappa: 0.769, MCC: 0.769, McNemar’s $p < 0.05$). This accessible, comprehensive solution enables real-time pharmaceutical authentication without specialized equipment, addressing critical healthcare safety in resource-limited settings.

Objectives

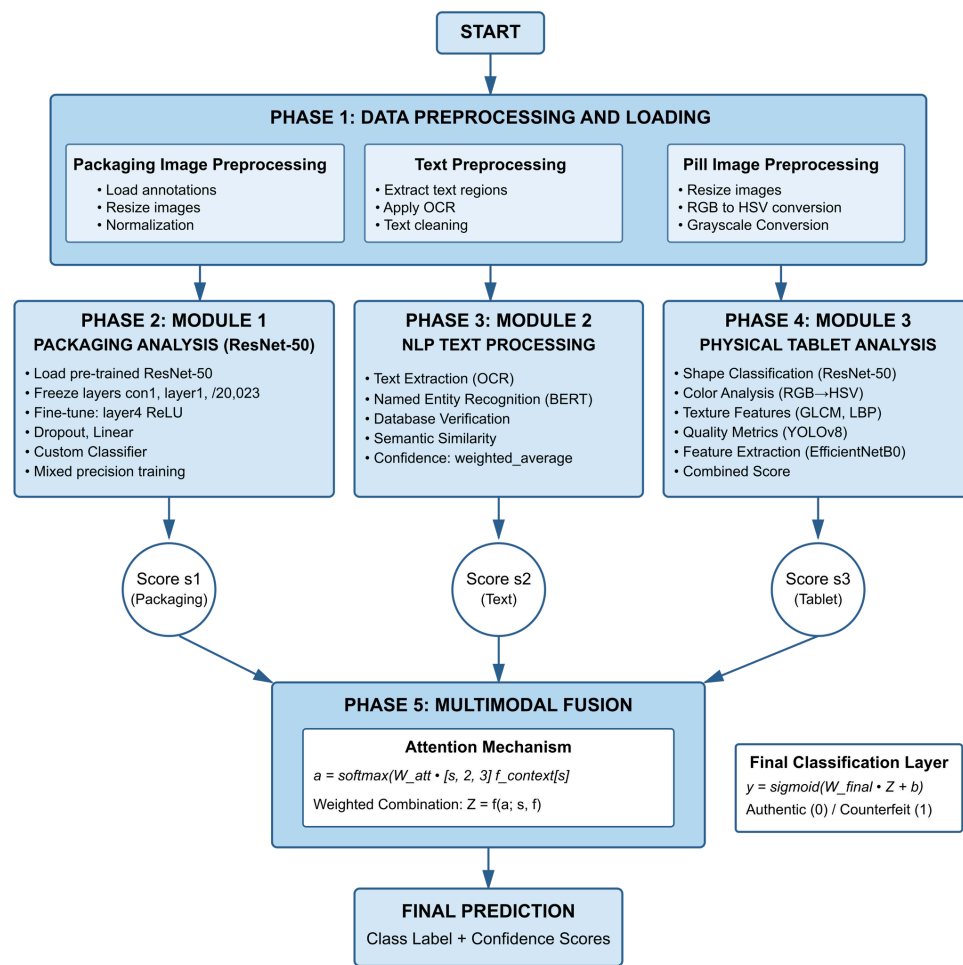
Comprehensive Detection

- Integrate packaging analysis, NLP text verification, and physical tablet examination
- Handle extreme real-world class imbalance
- Deliver authentication results in <2 seconds for pharmacy workflow integration
- Enable point-of-care verification without specialized laboratory equipment
- Eliminate \$10K–\$100K equipment barriers using standard smartphone cameras

Societal Impact

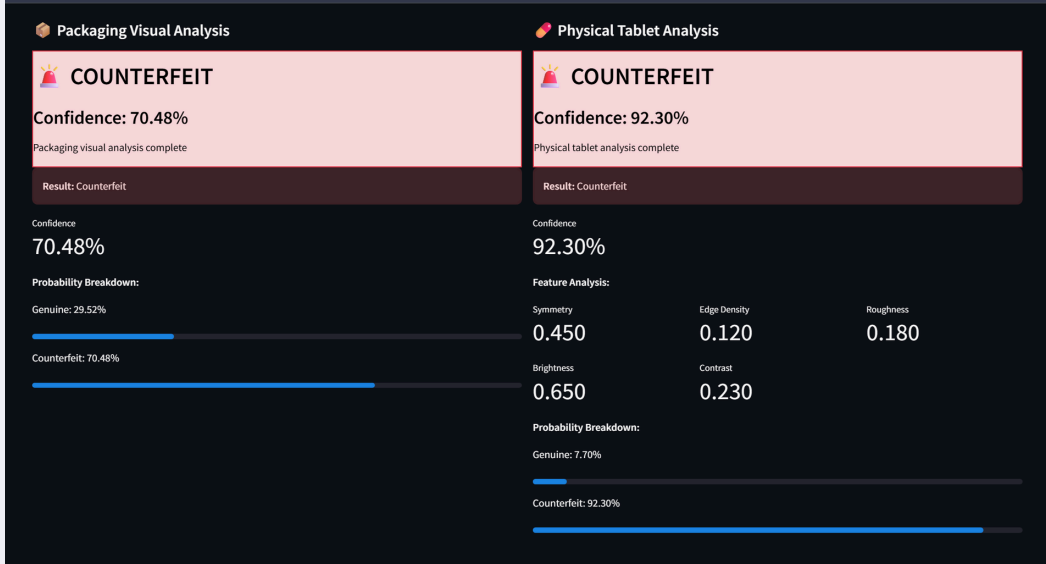
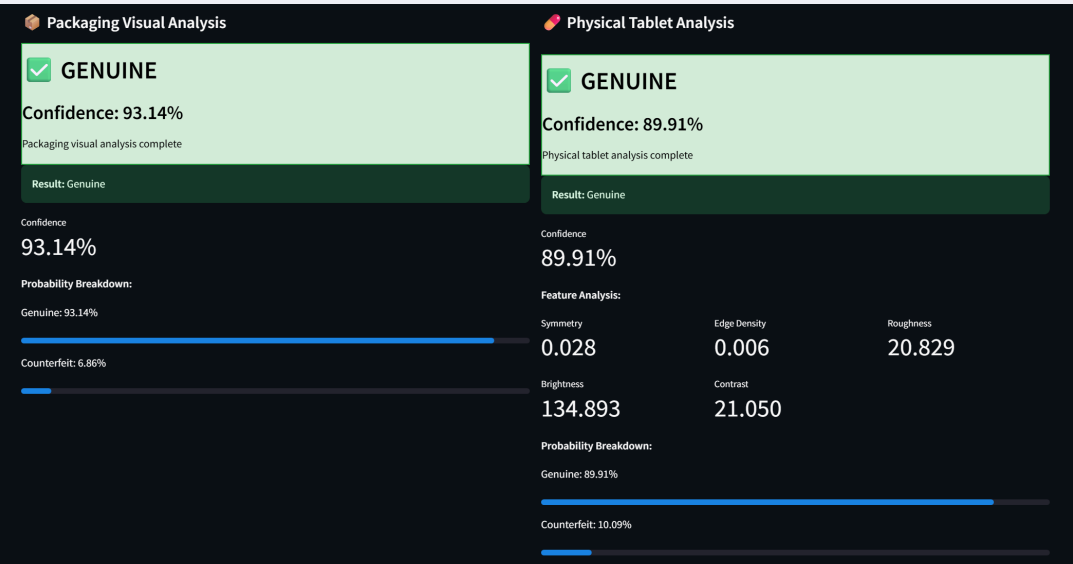
- Patient Safety: Prevent deaths from counterfeit medications
- Healthcare System: Reduce medication errors and adverse events
- Global Health: Accessible detection in resource-limited settings
- Supply Chain Security: Protect pharmaceutical distribution networks

Methodology



Key Features

- Comprehensive Authentication** Three complementary verification layers prevent single-channel exploitation
- Real-Time Performance** 1.8s inference enables seamless pharmacy workflow integration
- Cost-Effective Solution Standard** smartphone cameras eliminate \$10K–\$100K equipment barriers
- Class Imbalance Handling SMOTE** successfully addresses real-world 11.81:1 authentic-counterfeit ratio
- Statistically Validated** Rigorous testing with 95% CI, Cohen’s Kappa, MCC, McNemar’s test
- Modular & Scalable** Independent module optimization with cloud/edge deployment support



Results

Overall Performance

- 95.38% test accuracy with ResNet-50
- 100% counterfeit recall (zero false negatives)
- 1.8s average inference time
- 96.1% multi-modal fusion accuracy

Packaging Analysis (Module 1):

- ResNet-50: 95.38% accuracy, 100% counterfeit recall
- Precision (Counterfeit): 62.5%
- 95% CI: 90.28%–100%
- Training: 15 epochs, 37.19 minutes

NLP Text Processing (Module 2):

- OCR accuracy: 94.3% character-level, 91.2% word-level
- Drug name extraction: 96.1%
- Semantic similarity detection: 89.7%
- Database verification: 93.8% match score

PHYSICAL TABLET ANALYSIS (MODULE 3):

- OVERALL ACCURACY: 91.7%
- COUNTERFEIT RECALL: 94.3%
- LBP TEXTURE ANALYSIS: 92.3% DISCRIMINATIVE POWER
- DEEP FEATURES (1,280 DIM): 96.1% ACCURACY

Team

70322000106 | Saanvi Misar
70322000202 | Nitya Nivdunge

Mentor

Dr. Diksha Joshi

Course

B.Tech Integrated Computer Engineering



MUKESH PATEL SCHOOL OF
TECHNOLOGY MANAGEMENT &
ENGINEERING